

Sales Forecasting Report Using Time Series Analysis

The objective of this project is to analyze historical sales data and predict future sales using time series forecasting techniques. This helps businesses make informed decisions related to inventory management, marketing strategies, and operational planning.

Dataset Overview

- Dataset Name: Cleaned Synthetic E-commerce Sales Dataset.
- File Format: CSV.
- Number of Records: 100,000
- Number of Columns: 21

Key Columns Used:

- Order Date
- Revenue
- Month
- Year
- Season

Important Note:

The dataset contains transaction-level data which was aggregated into monthly sales data for forecasting purposes.

Time Series Data Preparation

Date Conversion

- The order_date column was converted into datetime format.

Data Aggregation

- Sales data was aggregated into monthly revenue to capture trends and seasonality.

Sorting and Indexing

- Data was sorted chronologically.
- The date column was set as the index for proper time series modeling.

Model Selection

Selected Model:

Holt-Winters Exponential Smoothing.

Reason for Selection:

- Captures both trend and seasonality.
- Suitable for business sales data.
- Easy to interpret and implement.

Alternative Models (Not Used):

- **ARIMA** – suitable for stationary data.
- **Prophet** – suitable for business forecasting with holidays.
- **LSTM** – requires complex deep learning setup.

Model Training and Testing

Data Split:

- 80% Training Data.
- 20% Testing Data.

Training Process:

- The model was trained using historical monthly sales data.
- Parameters used:
 - **Trend:** Additive.
 - **Seasonality:** Additive.
 - **Seasonal Period:** 12 months.

Forecasting Results

Test Predictions

- The model was used to predict sales for the test period.
- Predicted values were compared with actual sales.

Future Forecast

- Sales were forecasted for the next 12 months.
- The forecast reflects expected business trends and seasonal patterns.

Visualization Analysis

Actual vs Predicted Sales

- The model predictions closely follow the actual sales trend.
- Minor deviations are observed, which are normal in forecasting.

Historical vs Future Forecast

- The forecast shows a continuation of existing patterns.
- Seasonal peaks and dips are maintained in predictions.

Key Observations:

- Sales show consistent patterns over time.
- Seasonal fluctuations are clearly visible.
- No sudden or unrealistic spikes in forecast.

Model Evaluation

Evaluation Metrics:

- RMSE (Root Mean Squared Error)
Measures prediction error in actual units.
- MAPE (Mean Absolute Percentage Error)
Measures prediction accuracy in percentage.

Interpretation:

- Lower RMSE → better accuracy
- Lower MAPE → better forecasting performance

General Benchmark:

- MAPE < 10% → Excellent
- MAPE 10–20% → Good
- MAPE > 20% → Needs improvement

Key Insights

1. Sales exhibit a clear seasonal pattern across months.
2. Certain months consistently show higher revenue.
3. The business demonstrates stable growth trends.
4. Forecast indicates no major decline in future sales.
5. Seasonal peaks are expected to continue.

Business Implications

Increasing Sales Scenario

- Prepare higher inventory levels.
- Improve logistics and delivery systems.
- Increase workforce during peak periods.

Decreasing Sales Scenario

- Introduce promotional campaigns.
- Offer discounts during low-demand periods.
- Improve marketing strategies.

Seasonal Insights

- Plan festive offers in high-demand months.
- Optimize stock levels before peak seasons.

Business Recommendations

Inventory Management

- Maintain higher stock for peak months.
- Avoid overstocking during low-demand periods.

Marketing Strategy

- Run targeted campaigns during weak months.
- Maximize promotions during high-demand periods.

Demand Planning

- **Use forecast data for:**
 - supply chain planning.
 - warehouse management.
 - staffing decisions.

Strategic Planning

- Align business goals with forecast trends.
- Prepare for future growth opportunities.

Limitations

- Forecast is based on historical data patterns.
- External factors (economy, competition, policy changes) are not included.
- Accuracy may reduce for long-term forecasting.
- Dataset does not include special events like holidays or festivals.

Conclusion

The forecasting model successfully predicts future sales trends using historical data. The Holt-Winters model effectively captures both trend and seasonality, making it suitable for this dataset. The insights derived from forecasting can help businesses optimize inventory, improve marketing strategies, and enhance overall decision-making.

(Data Visualization charts & graphs are attached seperately)